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Bidirectional Coupling Simulation of Pressure-Saturation Decoupling to Enhance Fracturing-Flooding Performance in Low-Permeability Reservoirs

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Abstract

China's low-permeability reservoirs experience rapid production declines of 20-30% monthly, rendering conventional waterflooding methods ineffective due to inadequate injection pressures and limited sweep efficiency. To mitigate these issues, fracturing-flooding technology applies sustained high-pressure injections lasting 10-15 days above fracture gradients, facilitating enhanced fluid displacement through complex fracture networks. However, existing numerical simulation approaches struggle to accurately capture the dynamic evolution of reservoir properties during prolonged fracturing-flooding operations. This study introduces a bidirectional coupled modeling approach integrating geomechanical fracture analysis and reservoir fluid flow simulations. The framework simultaneously solves fluid flow and rock deformation equations, dynamically updating stress-dependent porosity and permeability, thereby providing improved predictive capabilities. Field validation shows the framework achieves less than 10% deviation in production history matching. The validated model reveals distinct propagation mechanisms: pressure fronts propagate rapidly, initially at 110 m/d and subsequently decreasing to 20 m/d, whereas saturation fronts advance at around 10.5 m/d, resulting in a clear velocity ratio of 2.2:1. Stress field analyses indicate notable reorientation of up to 40° near injection wells, contributing to complex fracture network formation. This modeling approach provides quantitative insights and practical guidance for optimizing fracturing-flooding operations, supporting improved management of pressure transmission and fluid displacement to enhance oil recovery in low-permeability reservoirs.

Keywords: Fracturing-flooding, Bidirectional coupling, Pressure-saturation propagation, Low-permeability reservoirs, Fracture network evolution

Introduction

The development of low-permeability reservoirs holds significant potential, accounting for approximately 38% of global petroleum resources, yet average recovery rates remain around 20%, leaving substantial

resources undeveloped (Wang et al., 2017; Chen et al., 2020; Wang et al., 2025). These reservoirs typically exhibit rapid production declines, and conventional recovery techniques are often economically unviable and operationally challenging, primarily due to their complex pore structures and high stress sensitivity (Li et al., 2021; Li et al., 2024; Lei et al., 2024). Consequently, improving the efficiency of low-permeability reservoir development has emerged as a critical challenge within the global energy industry (Wang et al., 2023).

Hydraulic fracturing technology is currently the primary method widely adopted to enhance single-well production in low-permeability reservoirs. However, this technology incurs high implementation costs, with fracturing fluids and proppants typically accounting for over 35% of total operation costs (Lu et al., 2023). Moreover, due to the relatively short duration of production enhancement effects and inability to provide long-term displacement energy, production gains typically decline rapidly (Liu et al., 2023; Bui et al., 2023; Wang et al., 2024). Meanwhile, conventional waterflooding technology, while capable of providing sustained displacement power, is limited by low injection pressures, making it difficult to form effective fracture networks and sufficient displacement pressure gradients in low-permeability reservoirs, resulting in injection-production difficulties and extremely low sweep efficiency (Settari, 1980; Soloveichik et al., 2022).

To address these technical bottlenecks, the recently emerged fracturing-flooding technology combines the fracture creation advantages of hydraulic fracturing with the sustained energy supplementation mechanism of waterflooding, proposing a new integrated development strategy (Lu et al., 2023; Su et al., 2025). Fracturing-flooding technology, through continuous high-pressure water injection lasting several days to over ten days, promotes long-term fracture network evolution while continuously and effectively providing reservoir pressure support (Guo et al., 2022; Yang et al., 2024). The distinguishing feature of this technology is complete water-based injection without requiring traditional fracturing proppants or additional fracturing fluids, thus offering significant cost advantages over conventional hydraulic fracturing. Through multi-field reconstruction and combined displacement and imbibition mechanisms, it can effectively enhance tight oil recovery (Lei et al., 2024). Field applications in multiple oilfields confirm that fracturing-flooding technology can significantly enhance oil production, with some pilot projects achieving daily oil production increases of up to 380% (Wang et al., 2025). Compared to traditional technologies, fracturing-flooding can establish a stable and effective "fracturing-displacement" coupling mechanism.

Recent years have seen extensive research on numerical simulation methods and mechanisms for fracturing-flooding. Cui et al. proposed a coupled simulation method combining Displacement Discontinuity Method (DDM) with black oil models to achieve simultaneous simulation of fracture propagation and reservoir flow (Cui et al. 2025). Zhao et al. established a parameter optimization model considering dynamic fracture propagation and proposed optimal injection schemes for fracturing-flooding (Zhao et al. 2025). Tang et al. studied the synergistic oil displacement mechanism of fracture propagation and surfactants (Tang et al. 2025). Su et al. employed Extended Finite Element Method (XFEM) with dual-porosity dual-permeability models to analyze the effects of in-situ stress and permeability on fracture propagation (Su et al. 2025). Despite these advances, existing studies generally neglect real-time interactive feedback mechanisms between fluid and rock deformation at field scale, failing to dynamically update reservoir parameters such as porosity and permeability in real-time, leading to error accumulation under long-term high-pressure injection conditions and significantly affecting simulation prediction accuracy.

Addressing these limitations, this study establishes a fully bidirectional coupled fluid-solid numerical simulation framework that simultaneously solves fluid flow and rock deformation processes while considering the impact of dynamic fracture conductivity evolution on reservoir flow characteristics. The framework achieves accurate description of fluid-rock interaction mechanisms during fracturing-flooding through real-time updates of stress-sensitive porosity and permeability parameters. The research focuses on revealing the differential physical mechanisms between pressure propagation and saturation advancement, quantitatively finding that pressure propagation velocity is approximately 2.2 times that of

saturation advancement, providing important theoretical basis for optimizing fracturing-flooding injection schemes. Through field data validation from representative low-permeability reservoirs in eastern China, the reliability and practicality of the bidirectional coupling framework in engineering applications is demonstrated, providing an effective numerical simulation tool for optimization design of fracturing-flooding technology in low-permeability reservoirs.

Methodology

Bidirectional Coupled Numerical Simulation Framework

This study establishes a fully bidirectional coupled numerical framework for simulating fracturing-flooding operations by integrating fluid flow dynamics with geomechanical deformation. The framework simultaneously solves the governing equations for fluid flow and rock mechanics to capture the interactions between pressure changes and stress evolution during reservoir stimulation and production.

The implementation process includes four key steps: (1) establishing the initial in-situ stress field distribution model and determining the geometric characteristics of initial fracture networks formed by natural fractures and hydraulic fracturing; (2) simulating fracture network propagation morphology during fracturing-flooding based on UFM fracture propagation model, employing LGR technology for refined grid characterization of fracture networks; (3) constructing multiphase flow numerical simulation model in complex fracture-pore media; (4) establishing real-time bidirectional coupling solution framework to quantitatively analyze flow propagation mechanisms and dynamic evolution of reservoir properties (porosity, permeability) during fracturing-flooding. The complete bidirectional coupling simulation framework is shown in Fig. 1.

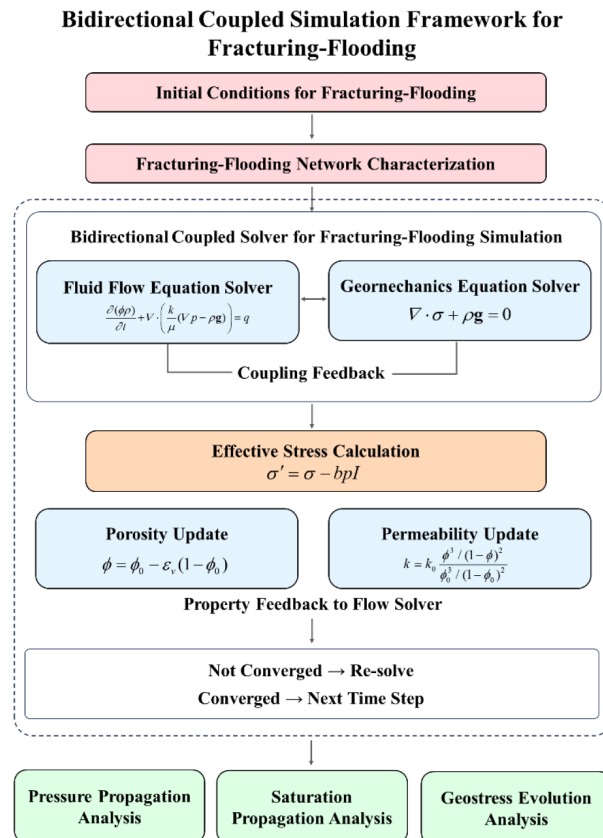


Figure 1—Bidirectional Coupled Simulation Framework for Fracturing-Flooding.

Governing Equations and Coupling Mechanism

Governing Equations. The bidirectional coupling framework solves the coupled system of reservoir fluid flow and rock deformation. The governing equations are:

Mechanical Equilibrium

$$\nabla \cdot \sigma + \rho \mathbf{g} = 0 \quad (1)$$

where σ is the total stress tensor, ρ is the density of the fluid-matrix mixture and $\mathbf{g}$ is the gravity acceleration vector.

Fluid Flow Equation

$$\frac{\partial(\phi\rho)}{\partial t} + \nabla \cdot \left(\frac{k}{\mu} (\nabla p - \rho \mathbf{g}) \right) = q \quad (2)$$

where ϕ is porosity, k is permeability tensor, μ is fluid viscosity, p is pore pressure, and q is the source/sink term.

Effective Stress Relationship

$$\sigma = \sigma' - b p I \quad (3)$$

where σ' is the effective stress tensor, p is the pore pressure, I is the second-order identity tensor and b is the Biot's coefficient.

Bidirectional Coupling Mechanism. The proposed fully bidirectional coupling mechanism enables real-time solution of fluid flow and rock mechanics interactions, specifically manifested in the mutual feedback between reservoir flow parameters and effective stress.

In forward coupling, fluid injection causes pore pressure changes, leading to effective stress redistribution and triggering rock deformation responses, ultimately causing dynamic changes in key physical parameters such as porosity, permeability, and fracture conductivity.

In backward coupling, dynamic changes in reservoir physical parameters alter fluid flow characteristics, further modifying spatial pore pressure distribution and causing secondary adjustments to reservoir rock stress states. Specifically, enhanced permeability in fracture zones increases fluid conductivity, accelerating pressure propagation and creating new stress concentration patterns, which in turn trigger additional permeability modifications. Enhanced porosity changes fluid storage capacity, affecting pressure buildup rates and subsequent stress evolution. This creates a continuous feedback loop where stress-induced property changes modify flow behavior, leading to pressure redistribution that generates new stress patterns and further property adjustments. This real-time interactive feedback mechanism effectively improves model prediction accuracy for actual reservoir dynamic responses.

Dynamic Updating Method for Porosity and Permeability

During forward coupling, reservoir properties evolve according to constitutive relationships. Porosity update follows:

$$\phi = \phi_0 - \varepsilon_v (1 - \phi_0) \quad (4)$$

where ϕ_0 is initial porosity, and ε_v is volumetric strain.

Permeability evolution incorporates pore structure changes through the Kozeny-Carman relationship:

$$k = k_0 \frac{\phi^3 / (1 - \phi)^2}{\phi_0^3 / (1 - \phi_0)^2} \quad (5)$$

where k_0 is initial permeability.

Real-time dynamic updates of porosity and permeability are achieved through continuous calculation of pore pressure and reservoir stress changes.

Numerical Model Setup

This study is based on actual geological parameters from typical reservoirs in eastern China, establishing a 3D reservoir model with dimensions of 2604 m×1688 m×523 m. Initial fracture networks have been accurately characterized in the model, with initial fracture zone permeability of 200 mD. Other key model parameters are shown in Table 1.

Table 1—Key Parameters for Fracturing-Flooding Simulation Framework.

Parameter	Value	Parameter	Value
Elastic modulus (GPa)	28	Pore pressure (MPa)	52.5
Poisson's ratio	0.28	Maximum horizontal principal stress (MPa)	80
Initial permeability (mD)	5.5	Minimum horizontal principal stress (MPa)	72
Injection rate (m³/d)	1000	Breakdown pressure (MPa)	58
Porosity (%)	10.8	Injection duration (days)	15

Model Validation

Field data validation employs actual production history data to verify the constructed bidirectional coupling model. Results show that prediction errors between modeled and actual production are below 10%, with pressure prediction errors controlled within 5%. Validation results confirm that the proposed bidirectional coupling numerical simulation model effectively captures complex dynamic characteristics exhibited by reservoirs during fracturing-flooding, demonstrating high simulation accuracy and good engineering applicability.

In summary, the proposed fully bidirectional coupling numerical simulation framework clearly reveals fluid-solid coupling mechanisms during fracturing-flooding. Through field history matching validation, its advantages in actual reservoir dynamic response simulation are further highlighted, demonstrating high theoretical value and broad engineering application prospects.

Case Study and Results

Model Validation

Fracture Propagation Morphology Validation. To validate the accuracy of the UFM fracturing-flooding fracture propagation model, simulated fracture geometric parameters were compared with microseismic monitoring data analysis. Fig. 2(a) shows fracture propagation morphology after fracturing-flooding, and Fig. 2(b) shows the 3D numerical simulation grid with unstructured mesh characterization capturing complex fracture geometry. Fig. 3(a) displays spatial distribution characteristics of microseismic events during water injection, with events mainly concentrated in elliptical regions around injection wells, different colors representing microseismic activity intensity at different injection volume stages. UFM simulation predicted fracture propagation extent shows good consistency with microseismic monitoring extent, with fracture length prediction errors controlled within 5%, as shown in Fig. 3(b-c), providing reliable fracture geometry for subsequent bidirectional coupling simulation.

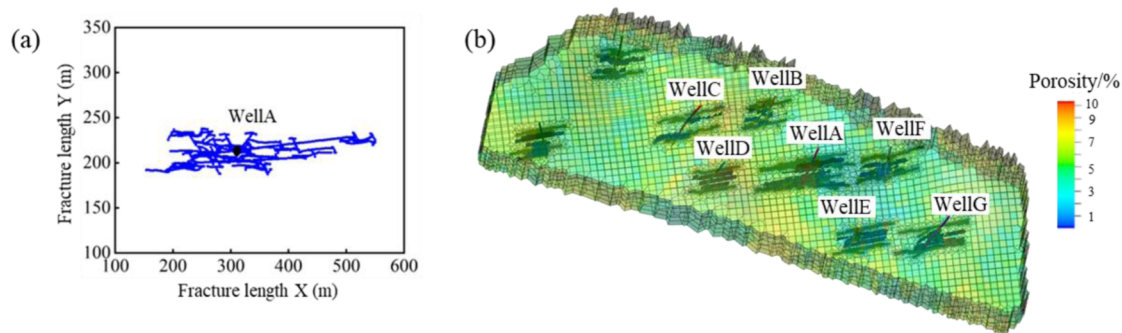


Figure 2—Fracturing-flooding induced fracture propagation and 3D numerical simulation mesh: (a) Planar view of fracture propagation; (b) 3D numerical simulation mesh for fracture propagation.

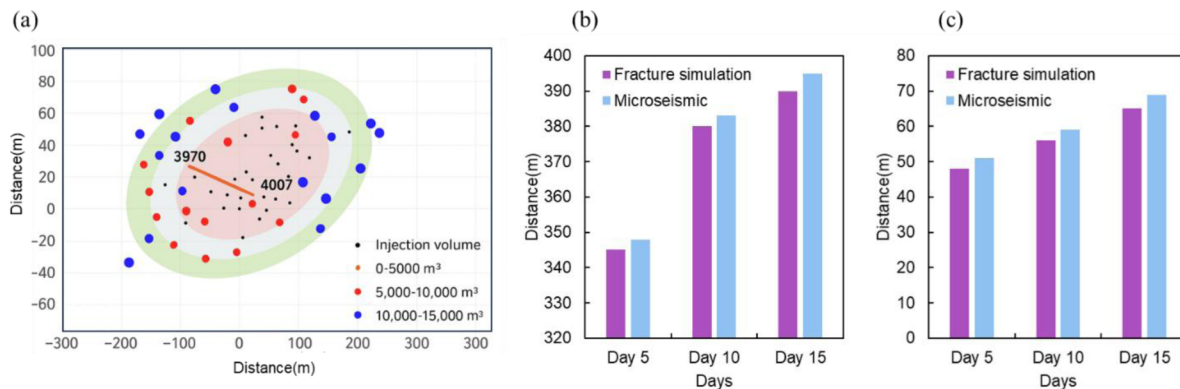


Figure 3—Microseismic monitoring comparison and fracture geometry validation: (a) Microseismic monitoring points after fracturing flooding; (b) Fracture lengths from simulation vs. microseismic data; (c) Fracture heights from simulation vs. microseismic data.

Production Performance History Matching. A comprehensive history matching analysis was performed on Well A group to quantitatively compare unidirectional and bidirectional coupling approaches under identical modeling conditions. The validation employed a liquid-constrained matching strategy, where total liquid production rates served as fixed boundary conditions. Oil production rates and water cuts were then systematically calibrated to maintain physical consistency among production parameters, while bottom-hole pressures (BHP) during injection periods were matched to verify pressure propagation accuracy.

The bidirectional coupling approach demonstrates enhanced accuracy compared to unidirectional coupling across all production metrics, with improvements becoming particularly pronounced during sustained high-pressure injection phases. Oil production matching accuracy improved from 89.1% to 94.2%, representing a 5.7% enhancement. Water cut predictions showed even greater improvement, increasing from 84.6% to 91.3% accuracy—a 7.9% gain. Most notably, bottom-hole pressure matching during fracturing-flooding operations achieved 96.8% accuracy with bidirectional coupling.

The enhanced performance of bidirectional coupling stems from its ability to capture real-time interactions between fluid flow and rock deformation that unidirectional coupling cannot account for. The framework captures stress-dependent permeability evolution in fracture zones, dynamic porosity changes affecting fluid storage capacity, and pressure-stress feedback loops where evolving flow properties influence pressure propagation patterns and create new stress distributions during high-pressure injection phases.

The superiority of bidirectional coupling was particularly evident during high-pressure injection phases, as shown in Fig. 4(c). During the November 1-16, 2020 injection period, pressure prediction errors remained within ± 0.15 MPa for bidirectional coupling, compared to ± 0.28 MPa for the unidirectional approach—nearly a 50% reduction in error magnitude. This consistent 6-8% accuracy improvement across all parameters demonstrates that bidirectional fluid-stress interactions become essential for reliable fracturing-

flooding simulation under sustained high-pressure conditions, providing the quantitative foundation necessary for optimizing injection strategies in field applications where dynamic coupling effects accumulate significantly over extended operation periods.

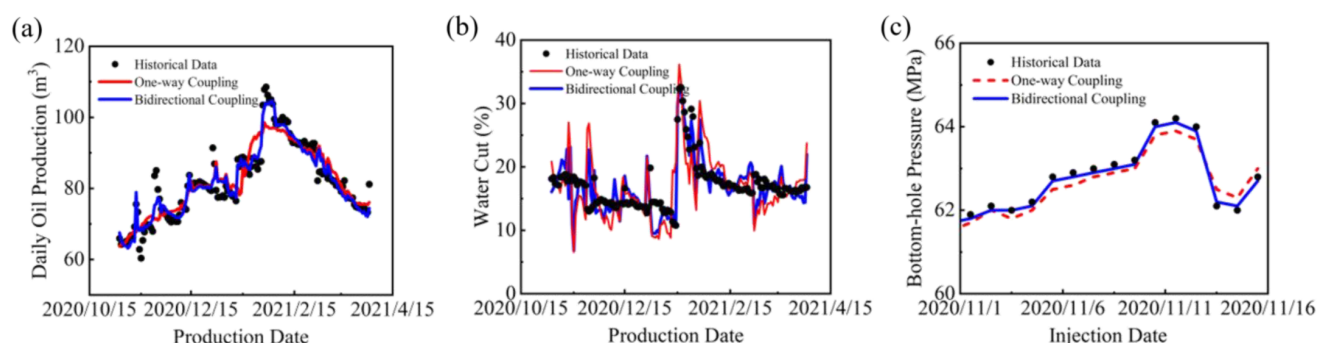


Figure 4—History matching comparison of production and injection performance: (a) Daily oil production; (b) Water cut; (c) Bottom-hole pressure during Fracturing-flooding.

Pressure Field Spatiotemporal Evolution

Following successful model validation, the framework is applied to analyze the spatiotemporal evolution of pressure and saturation fields during fracturing-flooding operations. To quantitatively distinguish between energy transmission and fluid displacement mechanisms during fracturing-flooding, explicit front advancement criteria are employed. Pressure front is defined as locations where pressure increases 1% above initial formation pressure of 52.5 MPa, while saturation front is defined as locations where water saturation increases 0.5% above initial value of 40.2%. This quantitative front definition method ensures consistency and comparability of propagation characteristic analysis at different times.

Reservoir pressure field exhibits significant spatiotemporal evolution characteristics during fracturing-flooding. Fig. 5 shows pressure distribution at different times: during early injection (Day 7), pressure disturbance mainly concentrates around injection well WellA, forming obvious high-pressure zones; as injection continues (Day 10), pressure influence range expands rapidly and begins affecting neighboring wells; during late injection (Day 15), effective pressure displacement system is established throughout the well group, with all production wells receiving pressure support.

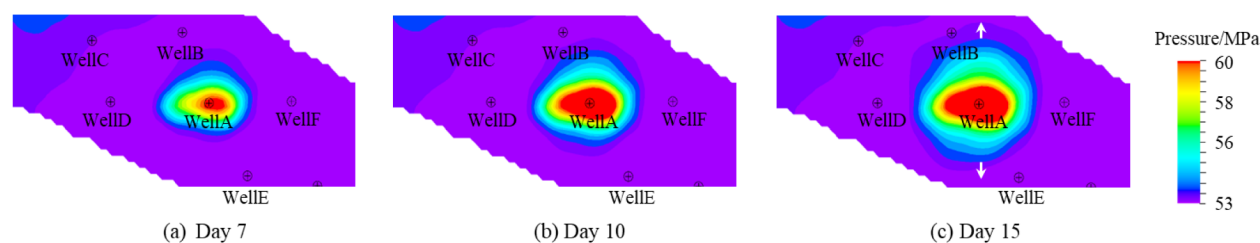


Figure 5—Formation pressure evolution during 15-day fracturing-flooding operation.

Quantitative analysis reveals that pressure front exhibits rapid expansion characteristics during the injection period. The pressure front equivalent radius grows from an initial 120m to 280m within the first 6 days, demonstrating exceptionally high initial propagation velocities of up to 110 m/d. This rapid expansion reflects efficient pressure conduction through the fracture network system. After 6 days, pressure propagation velocity decreases sharply to below 20 m/d as the expansion transitions from fracture-dominated to matrix-dominated propagation, ultimately reaching approximately 330m at 15 days.

Saturation Field Evolution

Compared to pressure propagation, saturation front advancement exhibits distinctly different characteristics. As shown in Fig. 6, injection fluid advancement is much slower than pressure propagation: at Day 7, water saturation increase is limited to near-wellbore regions of injection wells; at Day 15, saturation front remains mainly confined to fracture-developed zones, with limited matrix imbibition.

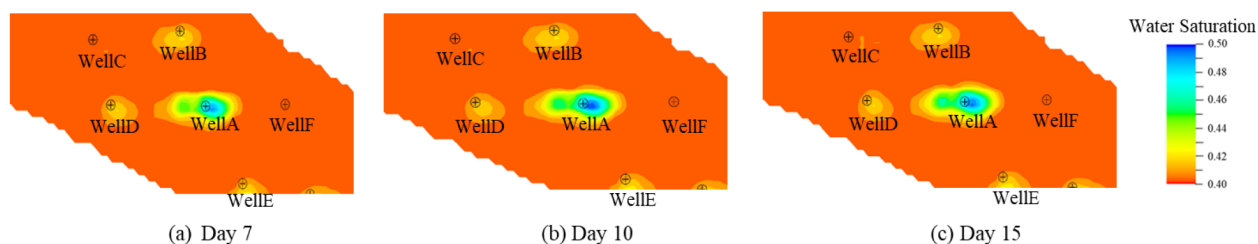


Figure 6—Water saturation evolution during 15-day fracturing-flooding operation.

Quantitative characterization of saturation front propagation reveals distinct transport behavior. The saturation front equivalent radius shows approximately linear growth, reaching only 150m at 15 days—less than half the pressure disturbance range. Saturation propagation velocity starts at approximately 100 m/d during initial injection but rapidly declines below 10 m/d after 3 days and remains stable thereafter. This rapid decay reflects preferential channel effects and capillary resistance in fracture-matrix systems.

Direct comparison of pressure and saturation front propagation reveals fundamental differences in transport mechanisms and provides quantitative evidence for the pressure-saturation decoupling phenomenon that characterizes fracturing-flooding effectiveness. As shown in Fig. 7, pressure front consistently leads saturation front by significant margins, with the equivalent radius ratio increasing from initial 1.2 to 2.2. The velocity comparison demonstrates that rapid pressure conduction through fractures enables extremely high initial propagation rates, while saturation advancement is limited by two-phase flow and capillary effects, with propagation velocity rapidly decaying to sustained lower levels.

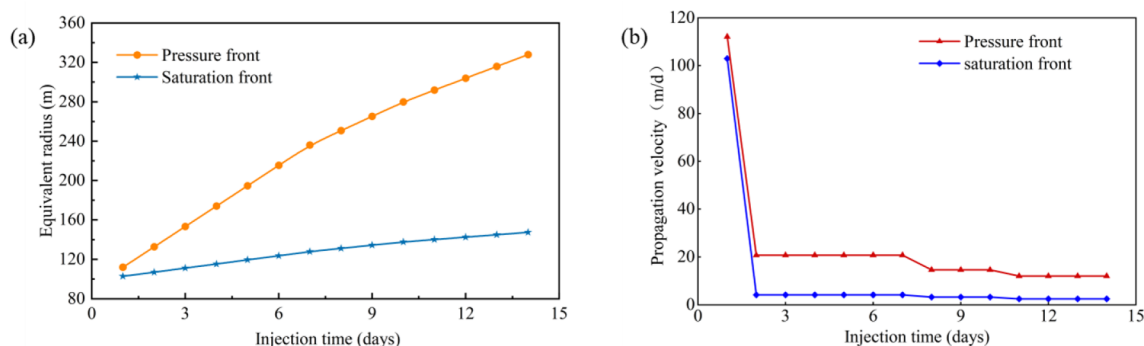


Figure 7—Pressure and saturation propagation dynamics during fracturing flooding: (a) Equivalent radius evolution showing greater spatial extent of pressure front compared to saturation front; (b) Propagation velocity comparison demonstrating rapid initial pressure transmission followed by sustained saturation advancement at lower velocities.

The significant velocity differential stems from fundamental differences in transport physics. Pressure propagation occurs through diffusive processes governed by formation compressibility, transmitting rapidly through both fractures and matrix at initial speeds exceeding 100 m/d. In contrast, saturation advancement requires actual fluid displacement through convective transport, constrained by relative permeability effects and capillary pressure barriers. This creates characteristic dual-porosity behavior where pressure equilibrates quickly between fractures and matrix, while fluid exchange remains limited by capillary imbibition rates. The low-permeability matrix (5.5 mD) further amplifies this decoupling, permitting

pressure diffusion but severely restricting convective flow, resulting in the observed 2.2:1 velocity ratio that provides quantitative design criteria for optimizing fracturing-flooding effectiveness in low-permeability reservoirs.

Stress Field Evolution

During fracturing-flooding, pore pressure changes from high-pressure fluid injection significantly alter reservoir stress field distribution. Stress field evolution characteristics after 15 days of injection reveal dynamic adjustment patterns of stress magnitude and orientation under fluid-solid coupling, as shown in Fig. 8.

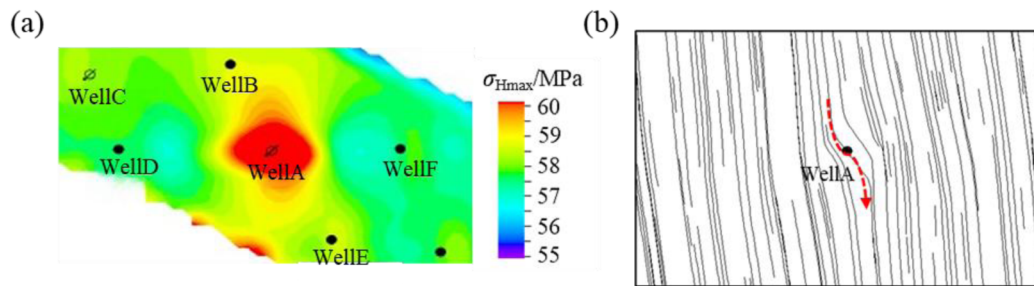


Figure 8—Stress field evolution induced by fracturing-flooding: (a) Minimum horizontal principal stress distribution; (b) Orientation changes of minimum horizontal principal stress.

Minimum horizontal principal stress distribution shows obvious stress concentration zones forming near injection well Well A, increasing from initial 56 MPa to 60 MPa after injection, with stress disturbance range reaching 200–300m, as shown in Fig. 8(a). This stress increase mainly results from poroelastic effects caused by high pore pressure. Rapid high-pressure injection forms high-stress zones centered at wellbores, gradually attenuating toward far-field and recovering to initial state, facilitating subsequent fracturing operations.

Minimum horizontal principal stress orientation shows significant deflection near Well A, with deflection angles reaching 30–40°, displaying radial distribution patterns, as shown in Fig. 8(b). This dramatic reorientation results from non-uniform rock deformation after fracture network formation: regions near fractures are compressed while inter-fracture regions expand, causing stress reorientation.

Maximum horizontal principal stress shows similar characteristics, with initial stress of 58 MPa increasing to 61 MPa after injection, as shown in Fig. 9(a). Stress difference decreases from 2 MPa to approximately 1 MPa, indicating stress field becomes more isotropic, favoring complex fracture network formation.

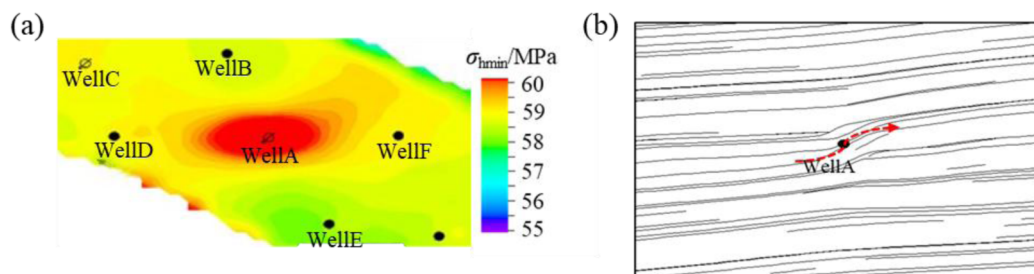


Figure 9—Stress field evolution induced by fracturing-flooding: (a) Maximum horizontal principal stress distribution; (b) Orientation changes of maximum horizontal principal stress.

Key impacts of stress field evolution include: first, high-stress zone formation favors subsequent fracturing; second, significant stress orientation deflection up to 40° promotes complex fracture network formation; third, reduced stress difference enables multi-directional fracture initiation.

Reservoir Property Evolution

Bidirectional coupling effects are fully manifested in spatially differentiated evolution of reservoir properties. Spatial distribution evolution of porosity and permeability multipliers exhibits distinctly different change patterns, as shown in Fig. 10.

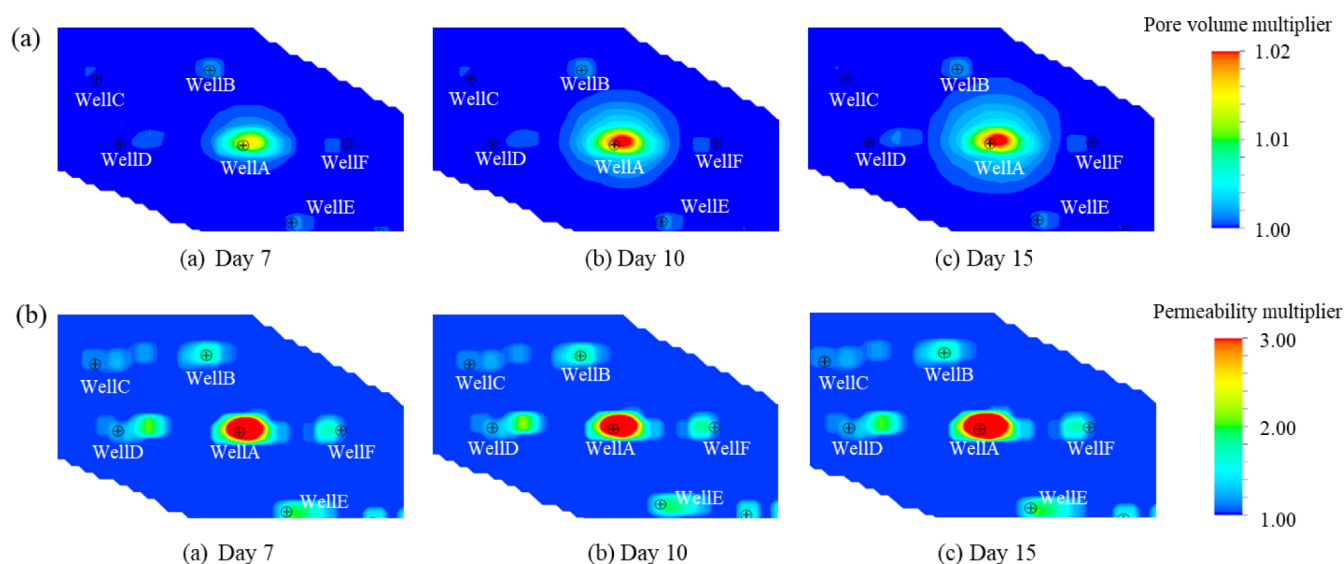


Figure 10—Spatiotemporal evolution of reservoir properties under bidirectional coupling effects: (a) Dynamic distribution of pore volume multiplier; (b) Dynamic distribution of permeability multiplier.

Porosity changes correlate closely with pressure influence range, with most significant increases in high-pressure regions, as shown in Fig. 10(a). Spatially, porosity increase zones show approximately circular expansion patterns centered at injection wells, consistent with pressure propagation fronts. After 15 days injection, porosity increase range exceeds 300m, with maximum increase approximately 2%. This pattern reflects the mechanical mechanism of pore pressure increase reducing effective stress, causing elastic expansion of rock framework.

Permeability evolution exhibits significant anisotropic characteristics, as shown in Fig. 10(b). Unlike uniform porosity expansion, permeability mainly increases along fracture directions, forming high-permeability bands near fractures. Permeability in fracture zones around injection wells can increase to 3 times initial values, while matrix regions distant from fractures show relatively limited changes. This spatial distribution indicates: first, fracture systems have higher stress sensitivity; second, effective stress reduction preferentially increases fracture aperture; third, fracture networks are primary channels for fluid flow and pressure transmission during fracturing-flooding. Preferential permeability increase further enhances fracture conductivity, creating positive feedback effects.

Discussion on Framework Capabilities and Limitations

While the bidirectional coupling framework successfully captures the complex hydro-mechanical interactions during fracturing-flooding, several limitations and challenges merit discussion. First, the current implementation assumes elastic rock deformation, which may not fully capture plastic deformation or fracture damage in highly stressed regions. Incorporating plasticity models would improve accuracy but significantly increase computational costs. Second, the framework currently models fractures as high-

permeability grid cells rather than discrete fracture networks (DFN), potentially oversimplifying fracture-matrix exchange processes in complex natural fracture systems.

Despite these limitations, this framework represents a significant advancement over existing approaches. Unlike conventional reservoir simulators that neglect geomechanical feedback, our bidirectional coupling captures the critical stress-permeability evolution that controls fracturing-flooding effectiveness. Compared to pure geomechanical models, our integrated approach enables direct production forecasting while maintaining mechanical accuracy. The key innovation lies in the real-time coupling of stress-dependent properties with multiphase flow, revealing the 2.2:1 pressure-saturation velocity ratio that provides quantitative design criteria previously unavailable to operators.

Future improvements could include: (1) implementation of embedded discrete fracture models (EDFM) for more accurate fracture representation, (2) incorporation of thermal effects for high-temperature reservoirs, and (3) integration of plastic deformation models to capture irreversible rock damage in highly stressed regions. These enhancements would broaden the framework's applicability while maintaining its core strength of capturing coupled hydro-mechanical processes essential for optimizing fracturing-flooding operations.

Conclusion

This study establishes a comprehensive modeling framework for fracturing-flooding fracture network simulation and flow analysis in low-permeability reservoirs, bridging the gap between detailed mechanistic analysis and practical engineering applications. The main conclusions are:

1. The modeling framework successfully addresses limitations of existing numerical simulation methods for fracturing-flooding operations by integrating detailed fracture mechanics analysis with practical reservoir engineering applications. The bidirectional coupling simulation, performed after explicit modeling of fracture distribution and properties, significantly enhances simulation accuracy compared to conventional numerical simulation methods.
2. Quantitative analysis reveals that pressure propagation consistently outpaces saturation advancement by approximately 2.2 times in both spatial extent and velocity during fracturing-flooding operations. This discovery provides quantitative evidence for the dual mechanism underlying fracturing-flooding effectiveness, where rapid energy transmission establishes favorable pressure gradients while sustained fluid displacement achieves enhanced oil recovery in low-permeability formations.
3. The validated framework achieves greater than 94% accuracy in production history matching for eastern China low-permeability reservoirs, demonstrating reliable predictive capability for oil production rates, water cut evolution, and bottom-hole pressure responses. Field validation confirms the framework's capacity to simulate fracturing-flooding operations and analyze pressure-saturation propagation mechanisms in low-permeability reservoir development applications.
4. Stress field evolution analysis reveals significant stress reorientation up to 40° near injection wells, with stress magnitude increases of 4-5 MPa creating favorable conditions for complex fracture network development. The bidirectional coupling mechanism captures the positive feedback between pressure-induced stress changes and permeability enhancement, with fracture zones exhibiting 3-fold permeability increases while porosity changes remain limited to 2%.

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